

SML

Assignment 4

April 2026

Instructions

- The assignment carries 6 marks. All questions and parts must be answered. The plots and final results should be included in the report. Support the plots and results with your analysis.
- Solve each question in a separate `.py` file. Add the `.ipynb` file if needed.
- Use <https://storage.googleapis.com/tensorflow/tf-keras-datasets/mnist.npz> MNIST dataset for the first two questions. Unpack the `.npz` file using `np.load()` to obtain the training set of size 60,000 and the test set of size 10,000. The preprocessing remains similar to the previous two assignments: flatten the images and normalize the pixel values.
- All experiments should be reproducible within the duration of the demo unless explicitly mentioned.

Question-1 [2 marks]

- Load the train and test sets for classes 4 and 9, and label them as -1 and 1 respectively. Divide the train set into train and validation set, by keeping aside 1000 samples from each class for validation. Apply PCA and reduce the dimensions to 5. For the remaining parts, use the reduced dimension dataset. Note that the validation set should not be used in obtaining the PCA matrix. [CO2]
- Grow a decision stump using the train set. For each dimension, the candidate split thresholds should be the midpoints between unique consecutive sorted values. You can randomly pick 1000 midpoints per dimension as candidate thresholds to speed up computation. Find the best split that minimizes the weighted misclassification error. Denote this as $h_1(x)$.
- Compute α_1 and update weights. [CO1]
- Build another tree $h_2(x)$ using the train set with update weights. Compute α_2 and update weights. Similarly grow 300 stumps. [CO3]

- Plot the accuracy on the validation set against the number of trees.
- Use the iteration that gives the highest validation accuracy and evaluate that iteration on the test set. Report test accuracy.

Question-2 [2 marks]

- Load the train and test sets for classes 4 and 9, and label them as -1 and 1 respectively. Divide the train set into train and validation set, by keeping aside 1000 samples from each class for validation. Apply PCA and reduce the dimensions to 5. For the remaining parts, use the reduced dimension dataset. Note that the validation set should not be used in obtaining the PCA matrix.
- Grow a decision stump using the train set. For each dimension, the candidate split thresholds should be the midpoints between unique consecutive sorted values. You can randomly pick 1000 midpoints per dimension as candidate thresholds to speed up computation. Find the best split that minimizes SSR. Denote this as $h_1(x)$. [CO2]
- For $\eta = 0.01$, compute residue as $y - \eta h_1(x)$. [CO1]
- Update the labels of the train set as done for absolute loss (obtained using negative gradients). Fit the next stump $h_2(x)$ on these updated labels. Now, compute residue as $y - \eta h_1(x) - \eta h_2(x)$.
- Similarly, grow 300 stumps and evaluate on the validation set after each iteration. Note that the labels are updated after every iteration. [CO3]
- Plot the validation MSE against the number of trees.
- Use the tree that gives the lowest MSE and report the test set MSE.
- Repeat the process with learning rates $\eta \in \{0.001, 0.01, 0.1, 0.2, 0.5, 1\}$. Comment on the trends and patterns observed with respect to the train, validation and test MSEs and the variation in the number of trees. (Not to be reproduced during the demo.) [CO1]

Question-3 [2 marks]

- **Dataset A:** Class -1 is sampled as $X \sim \mathcal{N}\left(\begin{pmatrix} -3 \\ -3 \end{pmatrix}, I\right)$ and Class 1 as $X \sim \mathcal{N}\left(\begin{pmatrix} 3 \\ 3 \end{pmatrix}, I\right)$. Obtain 200 samples from each class. [CO1]
- **Dataset B:** Sample classes with same means as Dataset A, but with covariance $3I$. Obtain 200 samples from each class.
- Use a 70 – 30 split for both datasets. Initialize all weights and biases to zero.

- Train the Rosenblatt perceptron from scratch without any library on Dataset A, with a learning rate $\eta = 0.01$ for a maximum of 300 epochs. Incorporate early stopping by halting training if no misclassification occurs during a full epoch. Plot the number of misclassified samples per epoch. Plot the final learned decision boundary overlaid on the training data as well as the validation data. Report convergence epoch (if observed) and test accuracy. [CO2, CO3]
- Repeat the previous step for Dataset B.
- Comment on the trends observed along with valid justification. [CO1]